



Pynance

Personal Cash Flow Management System

CS 4398 · Software Engineering Project

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A web-based finance tool focused on clarity, privacy, and simple day-to-day cash-flow tracking.

Problem & Motivation

Why we built Pynance

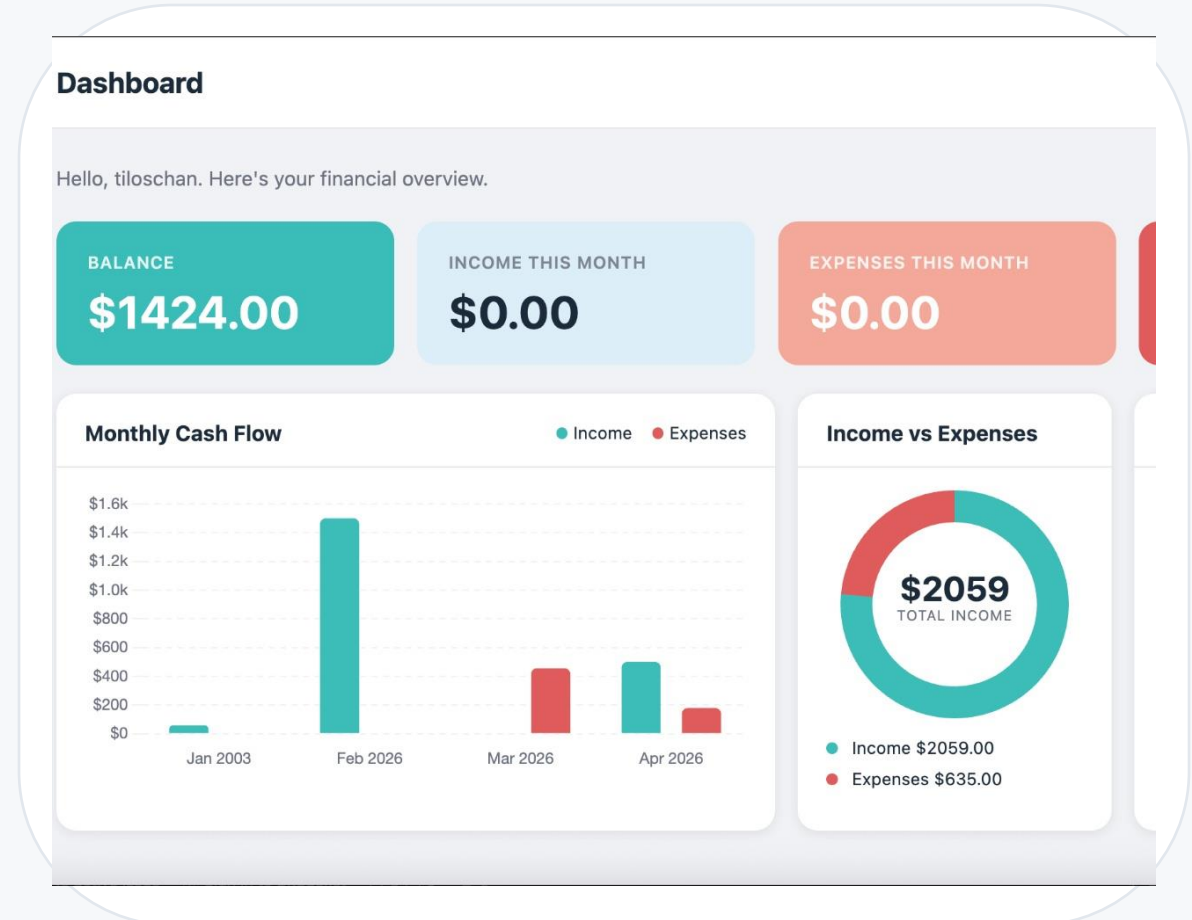
Many personal finance apps try to do too much at once.

Users often need a smaller, clearer tool that helps them answer simple but important questions: What came in, what went out, and am I moving toward my goals?

Complex interfaces Too many features distract from basic cash-flow understanding.

Manual awareness gap Users need to actively review spending patterns, not only see automated summaries.

Privacy expectations Financial records should stay tied to the authenticated user only.



Pynance focuses on clarity over complexity.

Solution Overview

A simple, private workflow for managing personal cash flow

Pynance gives users one place to record transactions, review financial summaries, visualize trends, and track savings goals.



Design principles

- **User control:** users manually add, edit, delete, and organize financial records.
- **Clear summaries:** dashboards convert transaction data into readable income, expense, and balance views.
- **Private scope:** records are connected to the logged-in user and are not shared across accounts.

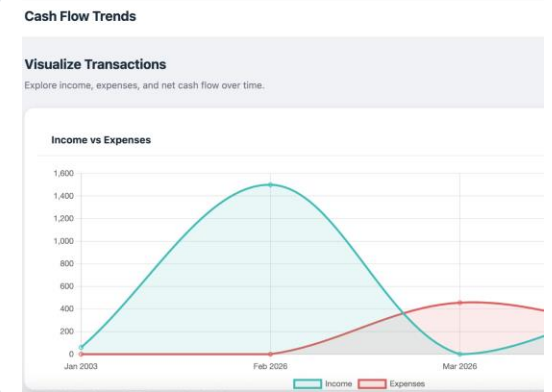
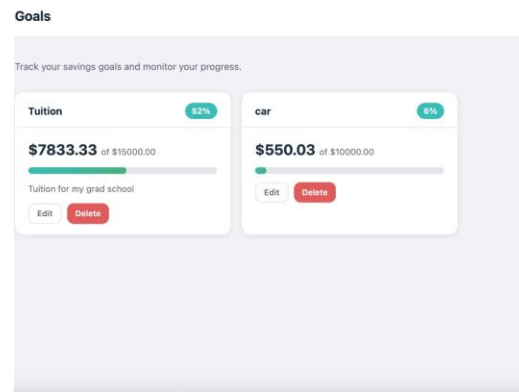
Core Features

What the application supports today

- **Authentication:** Register, log in, log out, and update account settings.
- **Transactions:** Create, edit, delete, search, filter, and categorize records.
- **Dashboard analytics:** Summarize balance, income, expenses, and recent activity.
- **Cash-flow charts:** Compare income and expenses across time.
- **Savings goals:** Create goals and track progress with percentage cards.
- **CSV support:** Import transaction data to reduce repetitive entry.

DATE	DESCRIPTION	TYPE	CATEGORY	AMOUNT	ACTIONS
April 21, 2026	First Salary	Income	Income	+\$500.00	Edit Delete
April 12, 2026	Rent	Expense	Variable Expense	-\$179.00	Edit Delete
March 5, 2026	Utilities	Expense	Fixed Expense	-\$456.00	Edit Delete
Feb. 1, 2026	Car sold money	Income	Income	+\$1500.00	Edit Delete
Jan. 1, 2003	—	Income	Income	+\$59.00	Edit Delete

Showing 5 transactions



Goals added as a core finance feature

User Workflow

From account access to financial action

The user flow was designed to be simple: get authenticated, enter the dashboard, manage financial records, then review the impact through charts and goals.

1

Account access

Users register or log in before reaching private financial pages.

2

Dashboard review

The system loads totals, recent transactions, charts, and navigation.

3

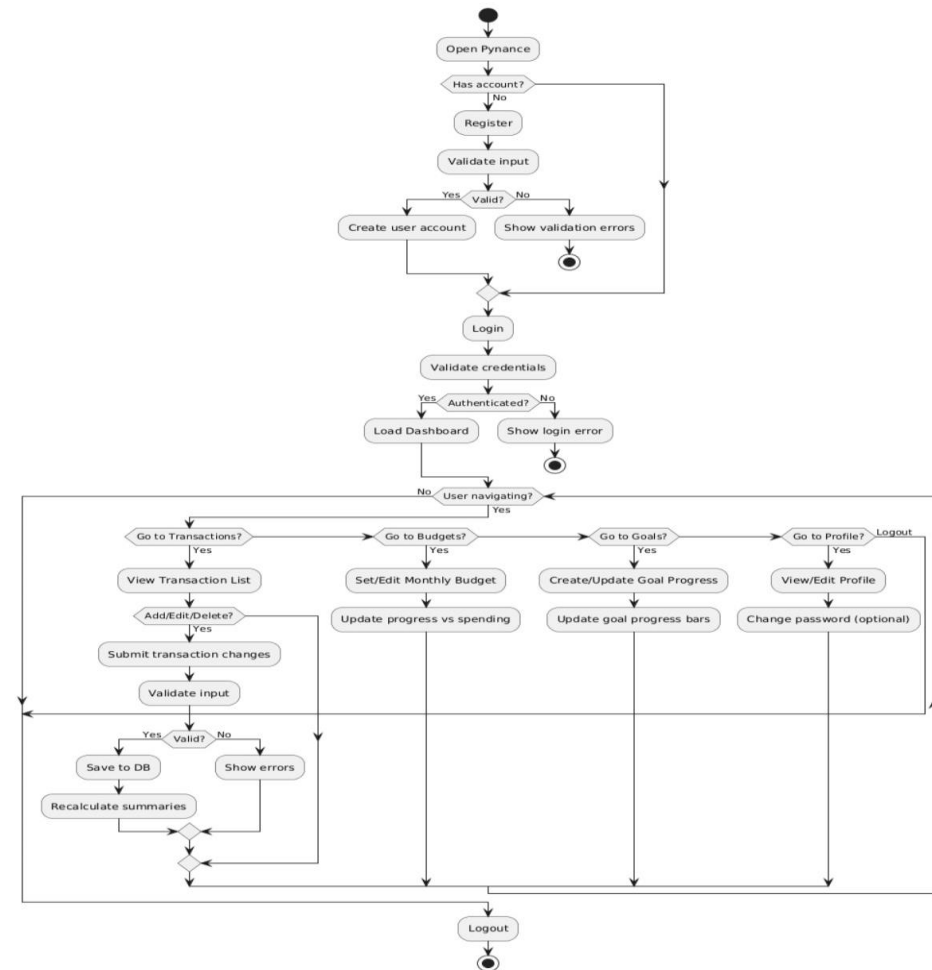
Manage records

Users add, edit, delete, search, and filter transactions.

4

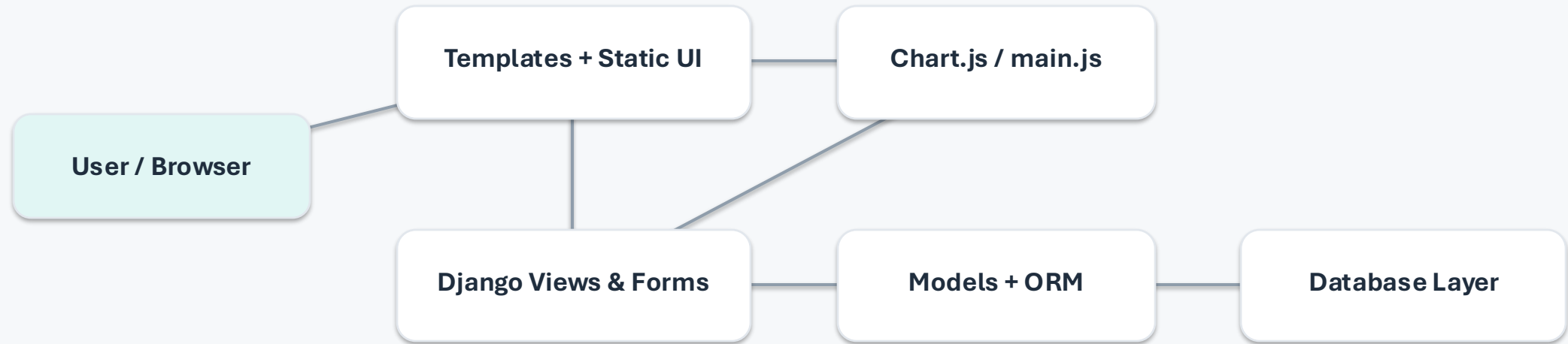
Track progress

Goals and cash-flow charts show progress over time.



System Architecture

Django-based application with template-driven frontend



Application modules

Accounts manages authentication and settings. Transactions handles CRUD and categorization. Dashboard aggregates financial totals and charts. Goals tracks target progress from user-entered data.

Data & Class Design

How the system organizes user financial information

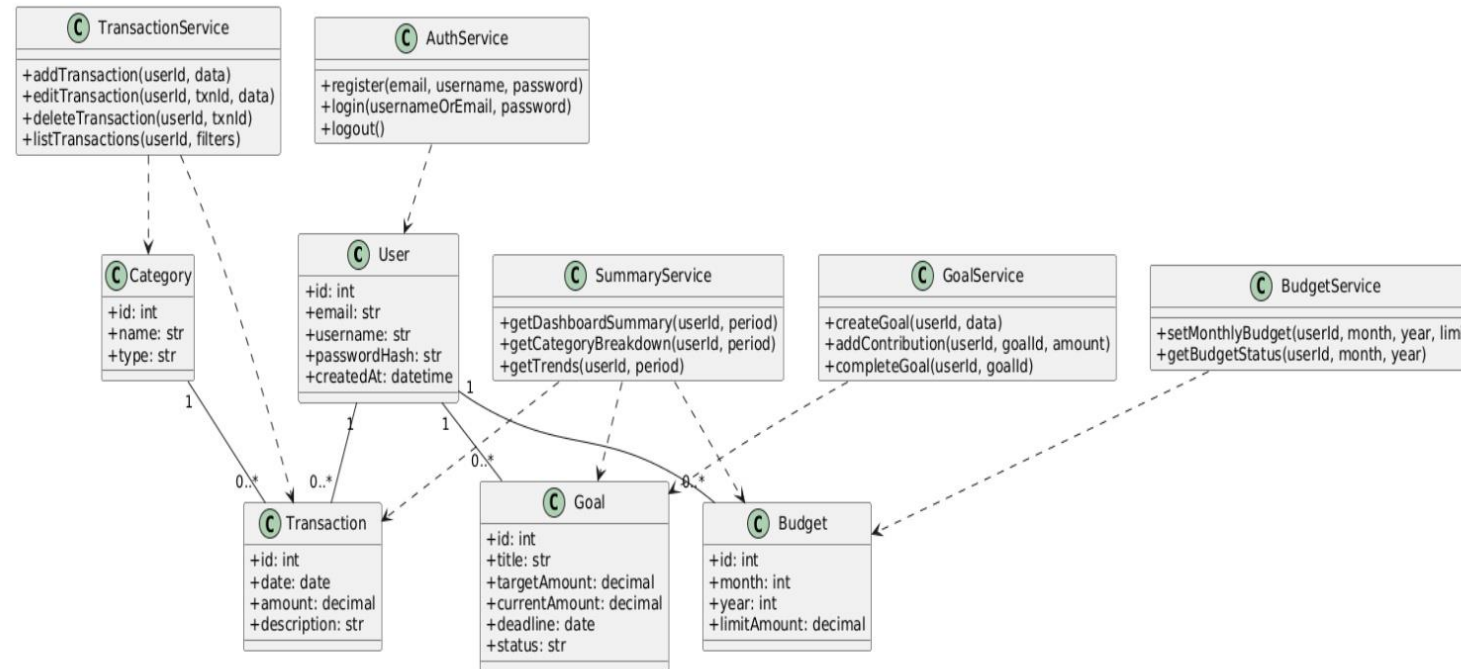


Figure 5: Class Diagram for Pynance

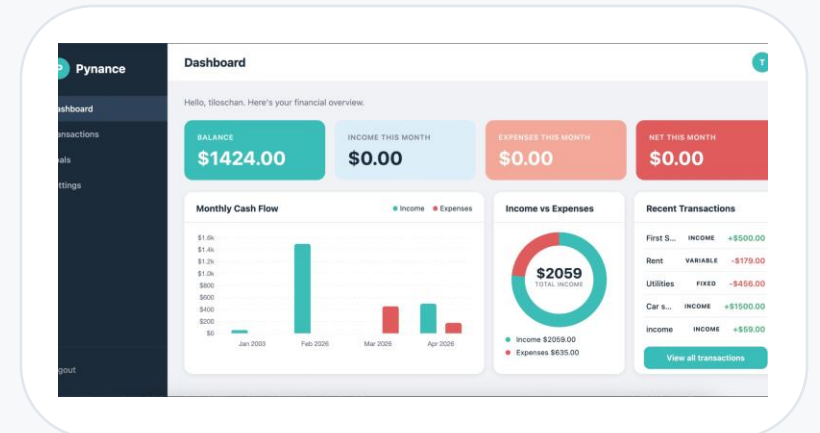
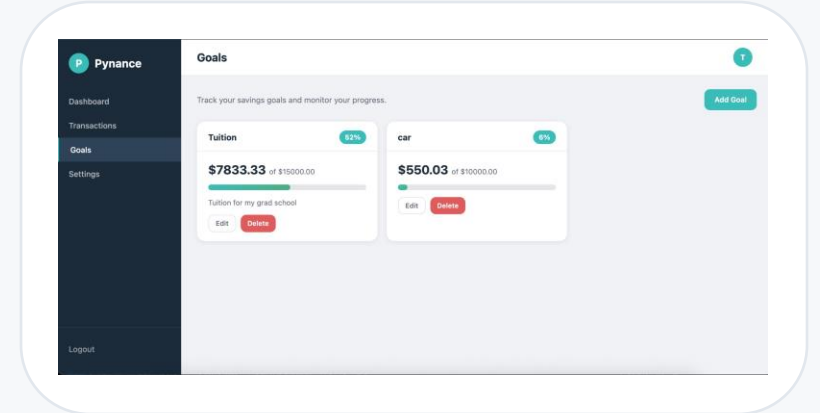
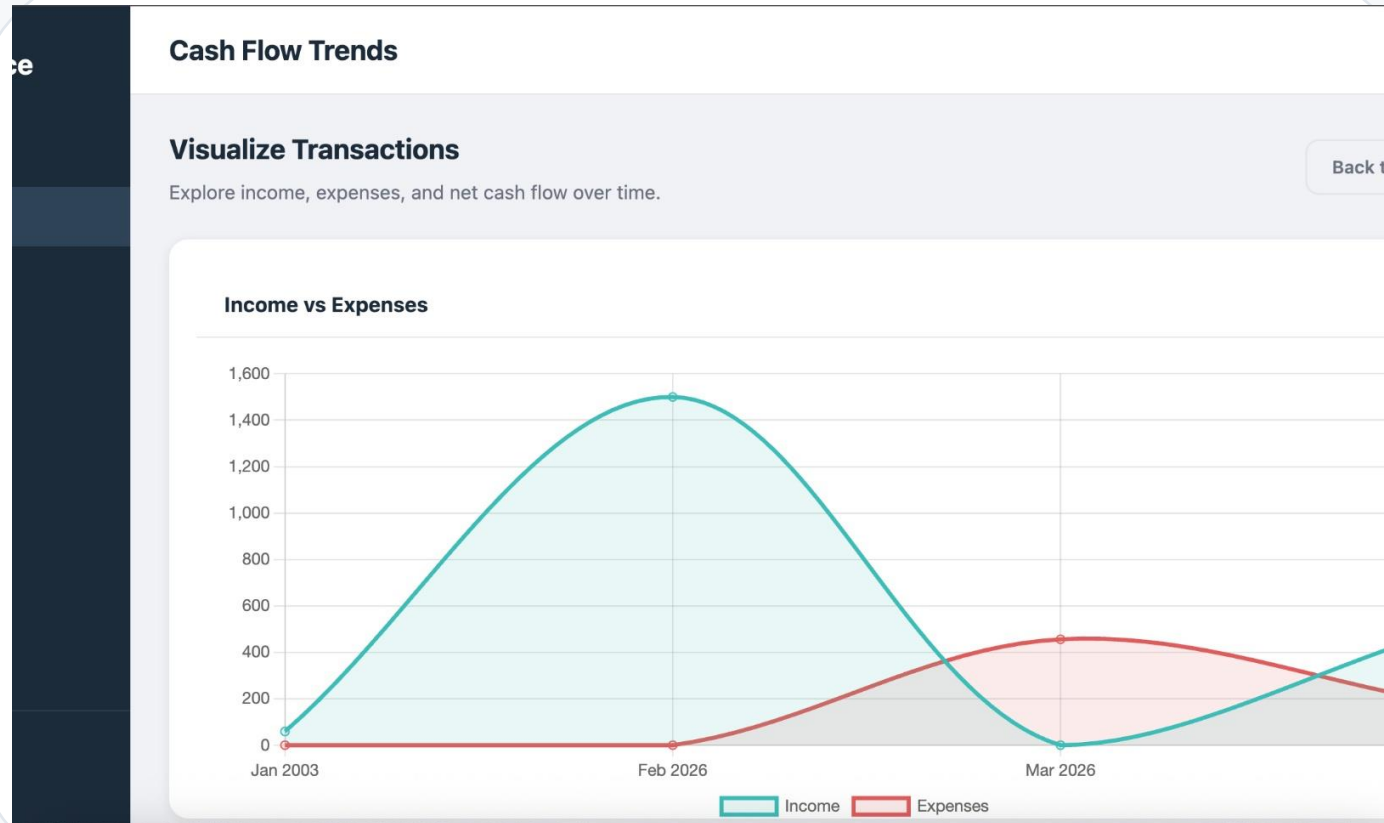
Design focus

- **User:** owns private transaction, budget, and goal records.
- **Transaction:** stores date, amount, category, and description for cash-flow tracking.
- **Category:** separates income, fixed expenses, and variable expenses.
- **Goal:** stores target amount, current amount, status, and progress.
- **Services:** represent core responsibilities such as authentication, summaries, transactions, goals, and budgets.

Key idea: user-scoped financial data

Analytics & Goal Tracking

Turning transaction data into useful financial feedback



Charts show income and expense movement over time, while goal cards translate saved amounts into progress percentages. Together, these features help users connect daily transactions to longer-term financial outcomes.

Prompt Engineering & Development Workflow

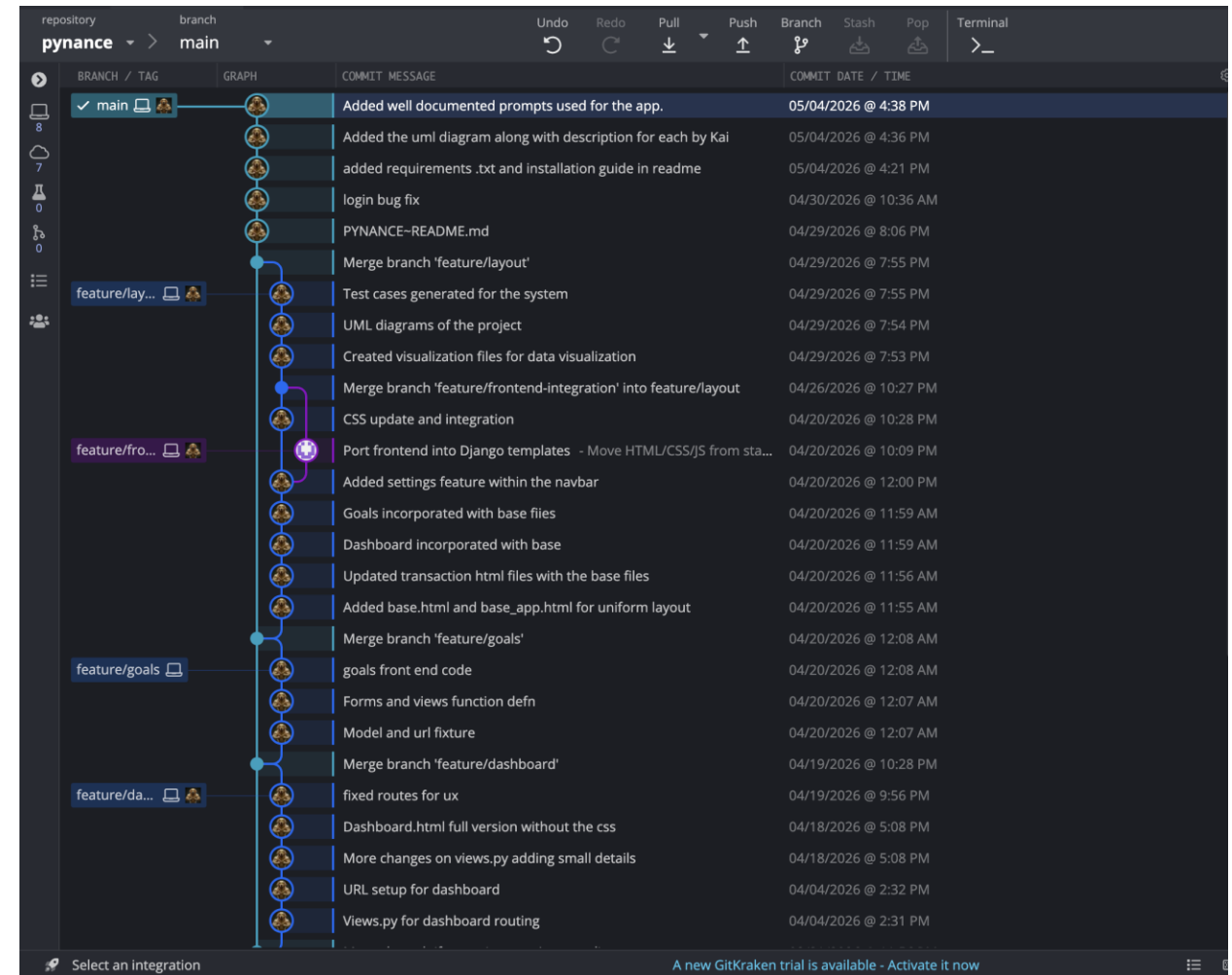
Using AI as an engineering assistant, not just a text generator

Prompt strategy

- **Feature prompts:** separate prompts were written for authentication, transactions, dashboard, goals, CSS, JavaScript, and documentation.
- **Implementation constraints:** prompts specified Django best practices, user data isolation, validation, and placeholder-free outputs.
- **Iterative debugging:** follow-up prompts targeted routing errors, static file loading, template issues, and UI scaling.

Result

The prompt documentation became part of the project record, showing how AI-supported development was planned, refined, and validated.



Version control shows feature branches merging into main, including frontend integration and layout work.

Acceptance Test Cases

Structured test coverage for authentication, transactions, and dashboard analytics

Authentication

TC-01: Register with valid input → Expected: Account created, redirect to login

TC-02: Login with wrong password → Expected: Error message displayed

TC-03: Verify logout ends session → Expected: Session cleared, redirect to login

TC-04: Access protected page → Expected: Redirect if unauthenticated

Total of 9 test cases in Authentication.

Transactions

TC-05: Create valid transaction → Expected: Record saved and listed

TC-06: Submit invalid input → Expected: Validation error shown

TC-07: Ensure users view only own records → Expected: No cross-user data

TC-08: Update own transaction → Expected: Changes persisted

TC-09: Delete own transaction → Expected: Record removed after confirm

Total of 11 test cases in Transactions.

Dashboard & Goals

TC-10: Verify income/expense/balance totals → Expected: Calculations correct

TC-11: Confirm chart data reflects transactions → Expected: Accurate visualization

TC-12: Create goal, add savings → Expected: Progress percentage updated

TC-13: Ensure only user-specific data shown → Expected: Data isolation confirmed

Total of 19 test cases in Dashboard & Goals combined.

Coverage Summary

39 total acceptance tests over authentication, transaction management, dashboard analytics, and goal tracking. Focus areas include data validation, user isolation, and correctness of financial calculations. All tests were manually validated with expected outcomes confirmed.

Challenges, Validation & Next Steps

Final project reflection

Challenges solved

- Static CSS files were not loading consistently across templates.
- Transaction routing and delete-confirmation behavior needed cleanup.
- The UI had to be restyled after removing Bootstrap.

Validation completed

- Confirmed login, registration, logout, and password update flows.
- Checked transaction add, edit, delete, search, and filter workflows.
- Verified dashboard totals, charts, recent transactions, and goal progress using sample data.

Future improvements

- Expand budgeting rules and goal recommendations.
- Improve CSV import feedback and error reporting.
- Add stronger automated tests and responsive mobile polish.

Conclusion

Pynance delivers a complete authenticated finance workflow: private user data, transaction management, dashboard analytics, visual cash-flow trends, account settings, and savings goal tracking. The system is clean enough for daily use and structured enough to grow into a fuller budgeting platform.